

Practical No:01

Aim: To create a database and collection using MongoDB commands.

Theory:

MongoDB is a NoSQL, document-oriented database management system designed for scalability, flexibility, and high performance. Unlike traditional relational databases that store data in rows and tables, MongoDB stores data in **JSON-like BSON (Binary JSON) documents**. These documents can hold nested structures, arrays, and a variety of data types, allowing developers to model real-world entities more naturally.

MongoDB follows a lazy-creation mechanism: a **database or collection is not physically created until data is inserted**. When a user switches to a database using the use command, MongoDB does not create the database immediately. Instead, it prepares the environment to create it only when the first document or collection is added. This behavior helps avoid unnecessary empty databases.

Collections can be created in two ways:

1. **Implicit creation** – A collection is automatically created when a document is inserted into it. This is the most common approach in MongoDB applications.
2. **Explicit creation** – Developers can explicitly create a collection using the `db.createCollection()` method. This method is particularly useful when advanced collection configurations are required. For example, administrators can create a **capped collection**, which has a fixed size and automatically overwrites old data when the collection reaches its size limit. Capped collections are beneficial for logging, caching, and real-time data streams.

In this experiment, the database `universityDB` is created implicitly when the user executes the `use universityDB` command. Then, the `students` collection is created explicitly using `db.createCollection("students")`. Once executed, MongoDB registers the collection inside the database, and it becomes visible when listing collections. This experiment helps students understand the basic database-creation workflow and the fundamental structure of MongoDB, which forms the foundation for CRUD operations, indexing, querying, and handling large datasets in future experiments.

Commands (mongosh):

```
use universityDB
```

```
db.createCollection("students")
```

Expected Output:

The `students` collection is created inside `universityDB`

Viva Questions:

- 1) How can a collection be created in MongoDB?

Answer:

A collection can be created implicitly by inserting documents or explicitly using the `db.createCollection()` command.

12. Reason
13. Remark

2) What is the purpose of `db.createCollection()`?

Answer:

It explicitly creates a collection and allows specifying advanced options like capped collections.

3) Which command is used to view all collections in a database?

Answer:

`show collections`

4) How is a database created in MongoDB?

Answer:

A database is created implicitly when data is inserted into it after using the `use databaseName` command.

5) What is a collection in MongoDB?

Answer:

A collection is a group of documents within a MongoDB database, similar to a table in a relational database.

Conclusion:

In this practical, we successfully learned how to create a database and a collection in MongoDB using basic MongoDB shell commands. The `use universityDB` command demonstrated MongoDB's lazy database creation mechanism, where the database is created only when data or collections are added. The `db.createCollection("students")` command explicitly created a collection within the database.